Politecnico di Milano

Embedded and Edge AI

TinyML Multi-Person Localisation from UWB Radar Signals

Phase-Aware Causal Inference on ESP32-S3 Constraints

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Selected model: V11, full-INT8 TensorFlow Lite

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This report documents the complete data analysis, preprocessing, model-selection, robustness and edge-optimisation pipeline. All numerical results are derived from the saved notebooks and machine-readable experiment artifacts included in the project repository.

Abstract

This project develops a TinyML system that estimates the two-dimensional positions of up to four people from raw complex UWB radar channel impulse responses. The input contains six radars, three antennas per radar and 120 range bins, while the target is an unordered set of person-presence and coordinate predictions inside a 4.8 m × 7.2 m room. The main challenge is to preserve angular, temporal and spatial information while satisfying an ESP32-S3 deployment budget.

The final representation combines log magnitude, a causal exponential-moving-average (EMA) residual, signed I/Q channels, stable inter-antenna phase differences and the immediately previous frame. A compact depthwise-separable convolutional network with 140,540 parameters produces four $[c, x, y]$ slots. Its metric-aligned, permutation-invariant training loss performs assignment and localisation in physical metres, matching the structure of the official Hungarian evaluation. On the selected group-disjoint validation split, the float model reaches 0.8516 F1, 0.8518 precision, 0.8514 recall and 0.430 m matched mean error. The deployed full-INT8 model retains 0.8339 F1 and occupies only 155.6 KiB. Its estimated activation arena is 103.9 KiB, it contains no recurrent, CUSTOM or Flex operations, and it passes the provided hard constraint screening. A two-fold acquisition-group assessment obtains 0.6867 pooled out-of-fold F1, exposing residual session sensitivity that is reported explicitly rather than hidden by the model-selection split.

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1 Problem definition and constraints

The objective is to infer a set of at most four two-dimensional person locations from one frame of raw UWB radar data. Each acquisition frame is represented as

$$\mathbf{R}_t \in \mathbb{R}^{6\times3\times120\times2},$$

where the final dimension contains the in-phase and quadrature components. The output contains four candidate slots,

$$\hat{\mathbf{Y}}_t = \{(\hat{c}_j, \hat{x}_j, \hat{y}_j)\}_{j=1}^{4},$$

with a confidence score and normalised coordinates for every slot. Active predictions are denormalised, clipped to the official room domain and written as JSONL.

The updated assignment defines four hard or practical constraints:

- the TFLite file must be smaller than 800 KiB;
- the estimated model activation arena must be below 300 KiB;
- LSTM, GRU, RNN, CUSTOM and Flex operations are forbidden by TensorFlow Lite for Microcontrollers (TFLM);
- the inference script may depend only on Python's standard library, NumPy, SciPy and TensorFlow.

Full INT8 quantisation is advisory rather than a formal hard requirement, but it is the natural choice for Flash, SRAM and execution efficiency on the ESP32-S3. The assignment does not require a physical microcontroller deployment or a C/C++ deliverable; deployability is assessed with the supplied constraint checker.

The official detection score first applies Hungarian assignment and accepts a pair only when the Euclidean distance is at most 1 m. Unmatched predictions are false positives and unmatched targets are false negatives. Consequently,

$$P = \frac{TP}{TP+FP}, \qquad R = \frac{TP}{TP+FN}, \qquad F_1 = \frac{2PR}{P+R}.$$

This definition makes confidence calibration, set assignment and localisation tails equally relevant to the final score.

2 Dataset and exploratory analysis

2.1 Dataset structure and integrity

The private `HAEEAI/multi-person-localization` dataset is distributed through Hugging Face as 24 acquisition windows. Each window contains 7,500 frames sampled at approximately 25 Hz, for a total of 180,000 frames and 420,000 valid person observations. Processing is performed one window at a time so that the complete raw radar tensor is never held in host memory.

All arrays have the expected dimensions. The EDA found no NaN or infinite radar values, no invalid person masks, no non-finite coordinates and no non-increasing timestamps. Generic imputation and timestamp repair were therefore unnecessary.

The occupancy distribution in Figure 1 is not uniform. Empty-room frames are the least frequent, whereas four-person scenes form the largest class. This motivated complete-window validation with explicit occupancy diagnostics rather than a single frame-averaged accuracy.

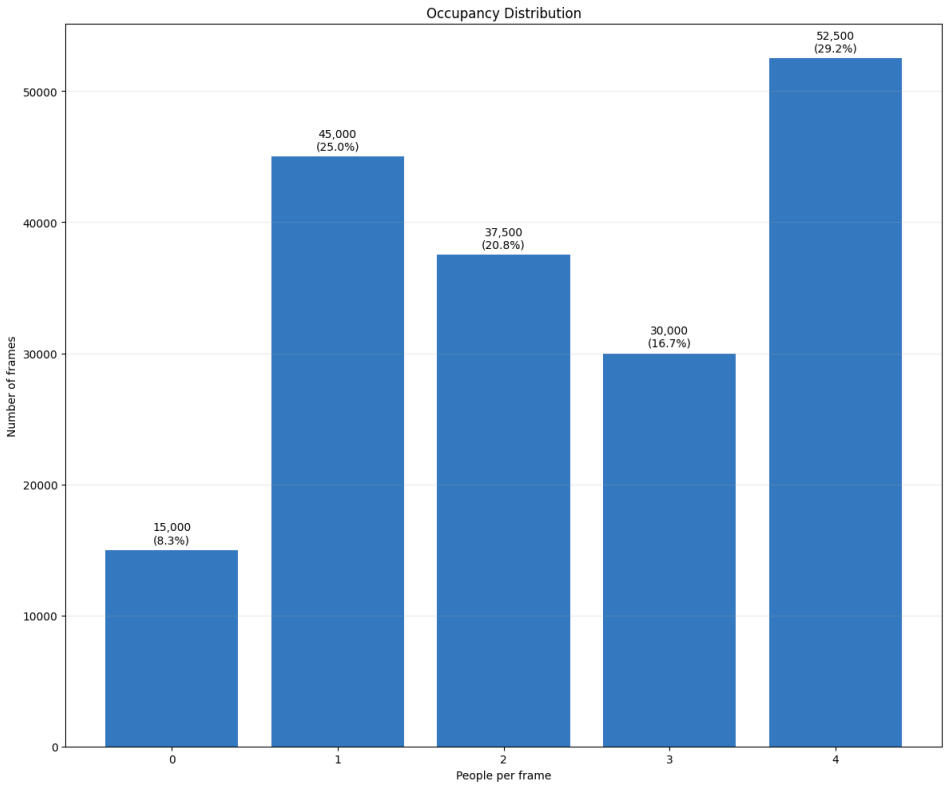


Figure 1: Occupancy composition of the 180,000 frames. Every level from zero to four people is represented, but the empty class accounts for only 8.3%.

2.2 Boundary observations and target policy

The EDA identified 1,081 coordinates outside the official room domain, corresponding to only 0.2574% of valid person observations. They affect seven windows and are concentrated mainly above the upper boundary. A temporal investigation found 16 contiguous segments, a median duration of 1.32 s and a maximum duration of 9.40 s. Their continuity indicates entrance and exit trajectories rather than independent annotation spikes.

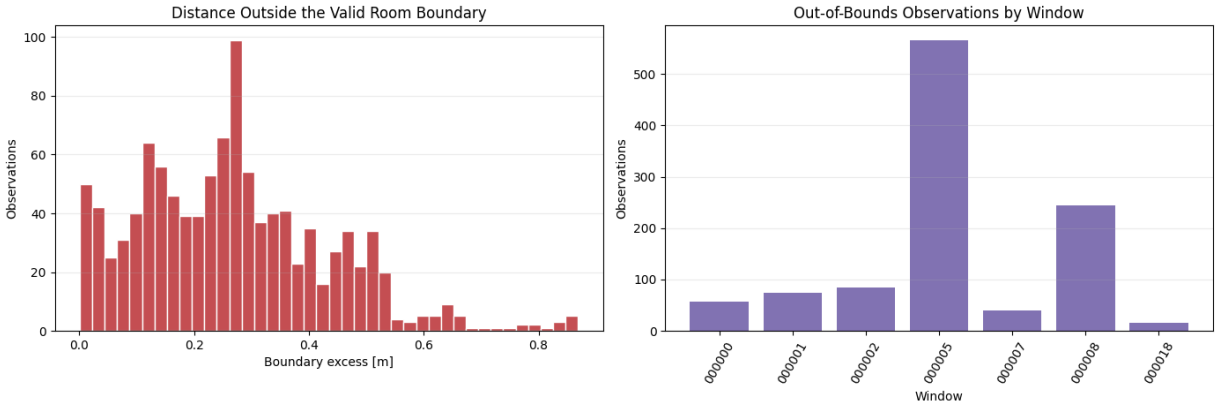


Figure 2: Magnitude and acquisition-window distribution of out-of-bound ground-truth observations. Removing all affected windows would discard far more valid data than the 0.2574% of coordinates requiring correction.

We retain the frames, person masks and counts, but clip valid target coordinates to $[0, 4.8] \times [0, 7.2]$ metres before normalisation. Final predictions are clipped to the same bounds. This policy

preserves scenario diversity and remains consistent with the evaluator, which rejects out-of-range predictions.

2.3 Acquisition-level shift and data splitting

Mean amplitude differs systematically between radars (approximately 1,247.9 to 1,358.4) and between acquisition windows (approximately 1,290.2 to 1,357.5). Adjacent frames within one recording are strongly correlated. A random frame split would therefore leak temporal and session information into validation.

All splits use complete acquisition windows. The primary split used for V1–V6 has 19 training windows and five validation windows. V7–V12 use a second, predeclared split with validation windows $\{22, 16, 2, 15, 7\}$, containing one complete window for each occupancy level. These are disjoint from the primary validation windows. The change was introduced to measure split sensitivity, not to search for a favourable score. Normalisation statistics are always recomputed on the training windows only.

3 Preprocessing and causal feature representation

3.1 Ground-truth representation

After clipping, valid coordinates are divided by the room dimensions. Initial experiments assigned people to four slots by lexicographic (x, y) order. Because the physical problem is an unordered set and slot order becomes unstable when people cross, later models use a permutation-invariant objective over all $4! = 24$ target assignments. Empty slots have confidence zero and zero coordinates.

3.2 Evolution of the radar representation

The preprocessing was developed through controlled stages:

1. **Magnitude:** $m = \log(1 + \sqrt{I^2 + Q^2})$, reducing the large CIR dynamic range while preserving range-bin energy.
2. **Signed I/Q:** $\operatorname{sign}(u) \log(1 + |u|)$, preserving quadrature cues discarded by magnitude alone.
3. **Fused magnitude and I/Q:** nine channels per radar, combining robust energy with signed complex information.
4. **Phase, background and temporal context:** the final 36-channel representation used by V10–V12.

For the final model, each radar has 18 current-frame base channels: three magnitude, three magnitude EMA residual, six signed I/Q and six phase-difference channels. For antenna pair (a, b), stable phase information is computed without angle wrapping:

$$\cos \Delta\phi_{ab} = \frac{I_a I_b + Q_a Q_b}{|z_a||z_b| + \epsilon}, \qquad \sin \Delta\phi_{ab} = \frac{Q_a I_b - I_a Q_b}{|z_a||z_b| + \epsilon}.$$

The pairs $(0, 1)$, $(1, 2)$ and $(0, 2)$ give six bounded channels. This exposes angle-of-arrival information using simple NumPy operations.

A causal background estimate is updated as

$$\mathbf{b}_t = \alpha \mathbf{b}_{t-1} + (1 - \alpha)\mathbf{m}_t, \qquad \alpha = 0.98,$$

and the residual uses $\mathbf{m}_t - \mathbf{b}_{t-1}$. The previous 18-channel base frame is concatenated with the current frame, producing the final $6 \times 36 \times 120$ tensor. Both states are reset only at acquisition boundaries, so inference remains causal and compatible with streaming batch-one operation.

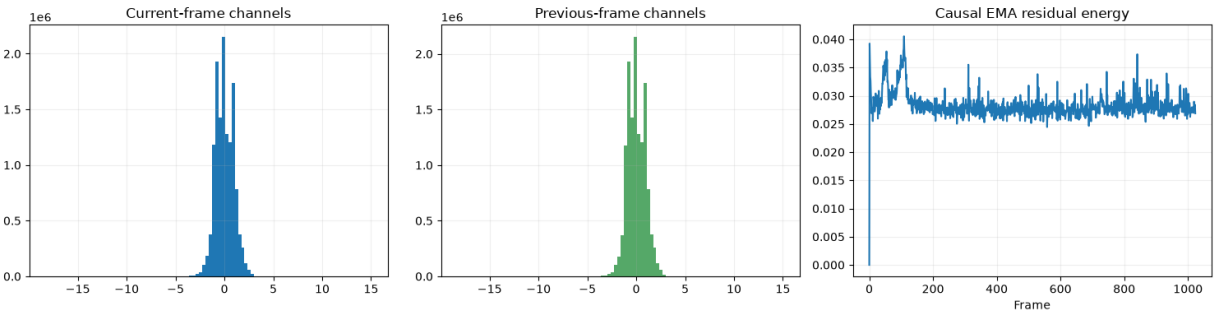


Figure 3: Validation diagnostics for current-frame features, previous-frame features and the causal EMA residual. The first-frame zero residual is the intended state initialisation, not a missing value.

Per-radar and per-channel means and standard deviations are estimated only from the training windows. The constants are stored in JSON artifacts and embedded in `submission/code.py`, preventing a train/inference mismatch. The implementation uses float32 preprocessing but quantises directly to the model's INT8 input scale.

4 TinyML model and metric-aligned objective

4.1 Architecture

The selected network is deliberately feed-forward and uses only standard TFLite operations. Table 1 summarises the deployed graph. The spatial layout is flattened after convolutional downsampling rather than globally averaged; this retains absolute radar and range-bin position, which proved important for localisation.

Table 1: V11 architecture. Batch normalisation is folded during TFLite conversion.

Stage	Configuration	Purpose
Input	$6 \times 36 \times 120$	Six radars, causal fused features
Range convolution	Conv2D, 24 filters, 1×7, stride 1×2	Early range-bin compression
Separable blocks	Depthwise 3×3, pointwise filters 24/32/48, ReLU, average pooling	Low-cost spatial feature extraction
Spatial head	Flatten, dropout 0.15, dense 64, ReLU	Preserve absolute spatial layout
Output	Dense 12 with sigmoid, reshape 4×3	Four confidence and coordinate slots
Total	140,540 parameters	

Depthwise-separable convolution reduces multiply-accumulate and parameter cost relative to a stack of full convolutions, following the efficiency principle used in MobileNet-style architectures [1]. Recurrent layers were intentionally avoided: one previous-frame feature state supplies short-term temporal information without unsupported LSTM, GRU or RNN operations.

4.2 Permutation-invariant physical-distance loss

Early models combined binary cross-entropy with coordinate-wise Smooth L1 loss. Although permutation-invariant matching improved slot stability, normalised x/y loss still treated the two room axes as if they had the same physical scale. V11 aligns training with the official metric by computing errors in metres.

For prediction slot j and a target under permutation π,

$$d_{j,\pi(j)} = \left\| \begin{bmatrix} 4.8 & 0 \\ 0 & 7.2 \end{bmatrix} (\hat{\mathbf{p}}_j - \mathbf{p}_{\pi(j)}) \right\|_2 .$$

The localisation term is a 1 m-delta Huber distance plus a continuous tail penalty $\max(0, d-1)$. The complete per-frame loss is

$$\mathcal{L} = \min_{\pi \in S_4} \left[\mathcal{L}_{\mathrm{BCE}}(\pi) + 2 \, \frac{\sum_j c_{\pi(j)} \left(\mathrm{Huber}_{1\mathrm{m}}(d_{j,\pi(j)}) + \max(0, d_{j,\pi(j)} - 1) \right)}{\sum_j c_j} \right].$$

The minimum over all assignments mirrors the unordered-set nature of Hungarian evaluation [2]. This loss is used only during training and therefore adds no operator, parameter or activation to the inference graph.

V11 was trained with Adam at an initial learning rate of 5×10^{-4}, batch size 128, at most 25 epochs, learning-rate reduction on plateau and restoration of the best validation-loss checkpoint. Fixed random seeds, window-level splits and saved normalisation artifacts support reproducibility.

5 Experimental progression and model selection

The notebook retains every experiment instead of overwriting unsuccessful versions. Table 2 reports the best threshold and global metrics measured in each experiment. V1–V6 and V7–V12 use different group-disjoint validation splits; comparisons across that boundary diagnose split sensitivity but are not controlled architecture comparisons.

Table 2: Complete experimental progression. P/R denote precision and recall.

V	Controlled change	Params	Thr.	P	R	F1	Count acc.
1	Minimal magnitude/global pooling	4,372	.85	.4419	.4329	.4374	.7109
2	Preserve spatial layout; set loss	134,996	.10	.4751	.5219	.4974	.7574
3	Spatial projection and regularisation	38,804	.55	.4226	.4638	.4423	.8037
4	Restored capacity; count calibration	134,996	.05	.4248	.4633	.4433	.8177
5	Signed I/Q representation	135,500	.30	.4759	.5213	.4976	.7895
6	Magnitude and signed-I/Q fusion	136,004	.30	.5229	.5297	.5263	.9432
7	Alternative acquisition split	136,004	.90	.4175	.4002	.4087	.9112
8	Radar-aware augmentation	136,004	.45	.4692	.4500	.4594	.8262
9	Intermittent energy-preserving augmentation	136,004	.65	.4470	.4467	.4469	.9718
10	Phase, EMA and previous-frame context	140,540	.75	.8108	.8109	.8108	.9950
11	Metric-aligned physical-distance loss	140,540	.65	.8518	.8514	**.8516**	.9846
12	AdamW and EMA weight averaging	140,540	.50	.8086	.8091	.8089	.8908

The first six experiments established that magnitude-only global pooling discarded important spatial information. Fusing signed I/Q produced the best primary-split F1 of 0.5263. V7 then revealed strong acquisition sensitivity on a second validation split. Augmentation in V8 and V9 improved some occupancy behaviours but did not solve localisation.

V10 produced the decisive gain from 0.4469 to 0.8108 F1. The improvement supports the physical hypothesis that phase differences expose angular information, while EMA and the immediately

previous frame describe changes over time. V11 changed only the loss and increased F1 by a further 0.0408 without increasing inference cost.

V12 tested AdamW with weight decay 10^{-4} and weight EMA momentum 0.99. It reduced F1 to 0.8089 and count accuracy to 0.8908, while matched error remained close to V11. Its degradation was therefore mainly confidence/count calibration, especially in the three-person window, rather than coordinate regression. V12 is retained as a negative experiment and was not converted to INT8.

6 Selected-model results and robustness

6.1 Validation convergence and occupancy behaviour

V11 reaches a best validation loss of 0.3944 at epoch 18. Figure 4 shows a smooth decline in training loss but substantial validation oscillation. The oscillation is consistent with a validation set made of only five complete sessions: each session is deliberately difficult and cannot be smoothed by frame-level mixing. Checkpoint restoration prevents the final epoch from being selected merely because of lower training loss.

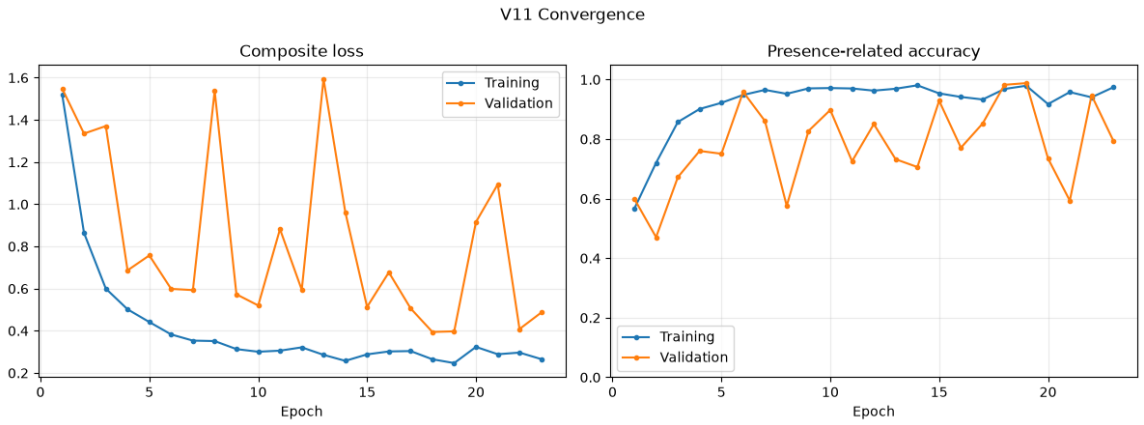


Figure 4: V11 convergence. Validation loss and presence-related accuracy are more volatile than training curves, reflecting acquisition-level rather than frame-level validation.

The optimal float confidence threshold is 0.65. At this operating point, precision and recall are balanced at approximately 0.852, count accuracy is 0.9846, matched MAE is 0.430 m and the 90th-percentile matched error is 0.790 m. Figure 5 also shows that one- and two-person windows are easiest, whereas the F1 decline in three- and four-person scenes remains the dominant weakness.

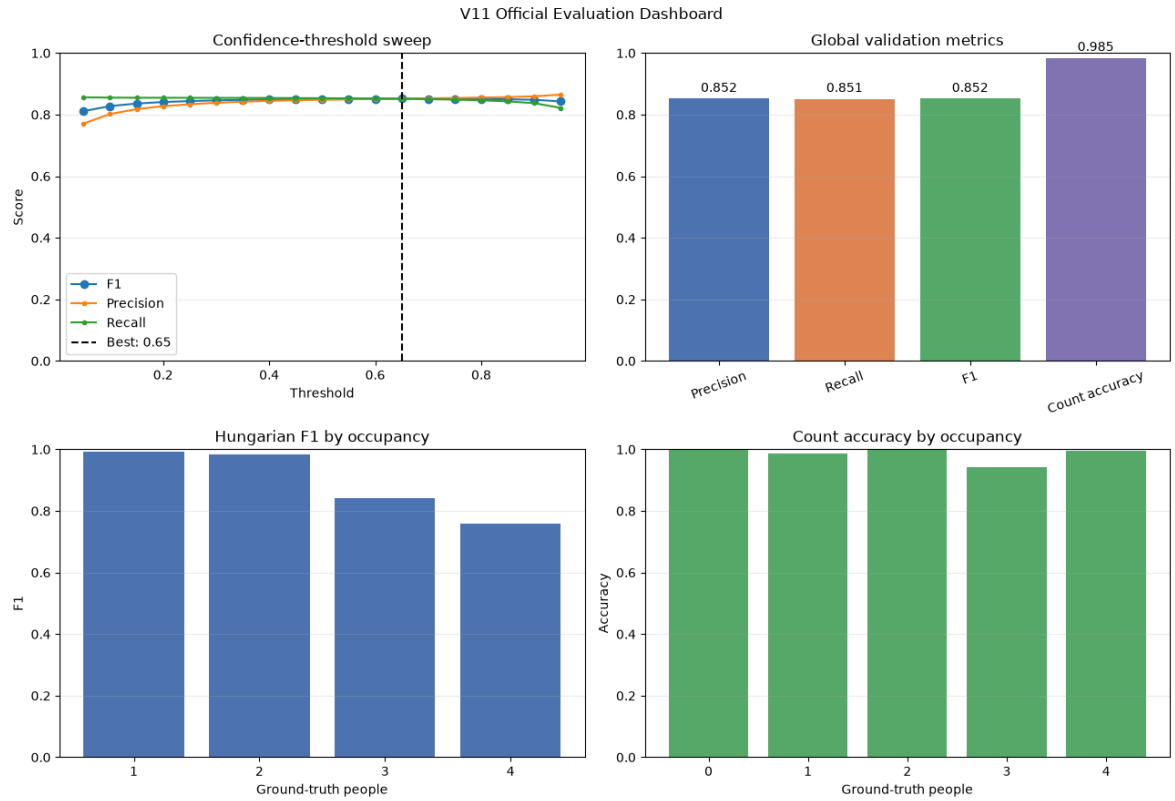


Figure 5: Official-style V11 evaluation: confidence sweep, global metrics, Hungarian F1 by occupancy and count accuracy by occupancy.

6.2 Acquisition-group cross-validation

A two-fold occupancy-stratified group protocol was used to quantify session sensitivity. Each complete acquisition appears in validation exactly once, all normalisation statistics are recomputed from the corresponding training fold, and the confidence threshold remains fixed at 0.65. Separate per-fold tuning would inflate the result.

Table 3: V11 group-disjoint two-fold robustness assessment.

	Frames	**Precision**	**Recall**	**F1**	**MAE (m)**	**Count acc.**
Fold 0	97,500	.6896	.7090	.6992	.531	.7735
Fold 1	82,500	.7354	.6146	.6696	.535	.6104
Pooled	180,000	.7077	.6668	**.6867**	.533	.6987

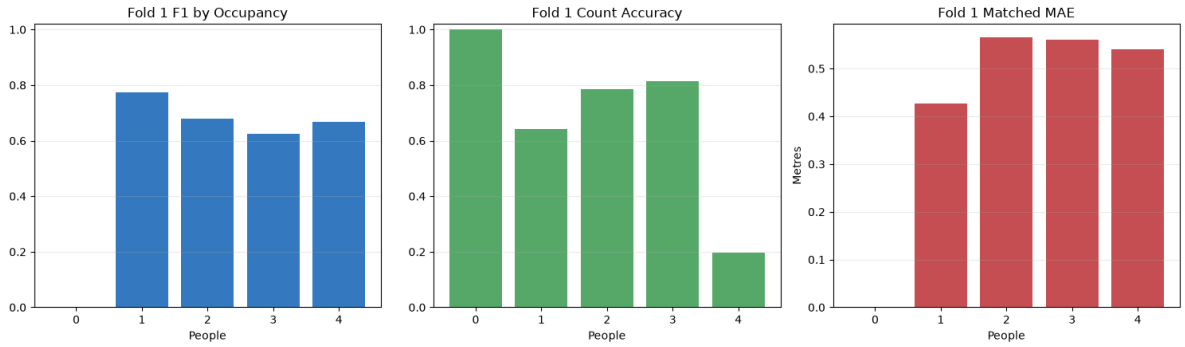


Figure 6: Fold-1 occupancy diagnostics. Four-person count accuracy is the clearest cross-session failure, even though empty-room rejection remains excellent.

The pooled F1 of 0.6867 is lower than the selected-split score of 0.8516. This is not a reason to discard cross-validation: it is evidence that the dataset contains strong session-specific variation and that the selected-split metric should not be presented as a guaranteed hidden-test score. The final evaluator uses another acquisition session, so this uncertainty is a central limitation of the system.

7 INT8 conversion and edge optimisation

7.1 Quantisation result

V11 was converted with a representative training dataset to a full-INT8 TFLite model with INT8 inputs and outputs. The selected deployed operating threshold is 0.75. The conversion retains 0.8339 F1, compared with 0.8516 for float inference, an absolute decrease of 0.0177. Matched MAE changes from 0.430 m to approximately 0.453 m, and count accuracy changes from 0.9846 to 0.9705. This was accepted as a favourable accuracy/edge-cost trade-off.

The model contains only `CONV_2D`, `DEPTHWISE_CONV_2D`, `AVERAGE_POOL_2D`, `FULLY_CONNECTED`, `LOGISTIC`, reshape, transpose, pack, shape and strided-slice operations. No forbidden recurrent, CUSTOM or Flex operation was found.

7.2 Flash, activation memory and latency

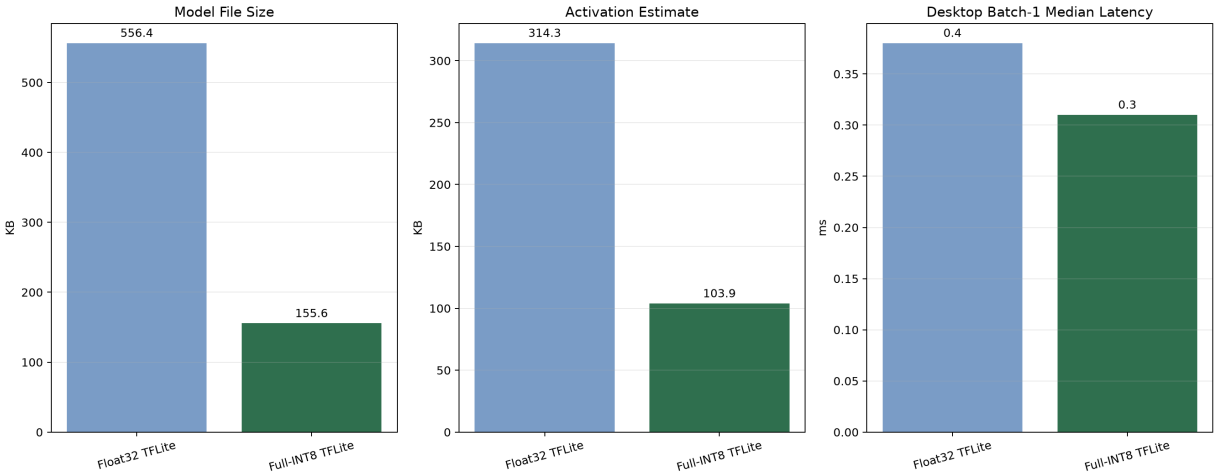


Figure 7: Float32 versus full-INT8 edge audit. Desktop batch-one latency is included only as a relative host measurement and is not an ESP32-S3 latency claim.

Table 4: Measured edge audit. KiB values use 1,024 bytes.

	Float32 TFLite	**Full INT8**	**INT8 improvement**
Model file	556.4 KiB	155.6 KiB	72.0% smaller
Activation estimate	314.3 KiB	103.9 KiB	66.9% smaller
Desktop median, batch 1	0.380 ms	0.310 ms	18.4% lower
Forbidden operations	none	none	–

The full-INT8 model leaves approximately 644.4 KiB below the Flash limit and 196.1 KiB below the official model-arena limit. The float activation estimate would exceed the updated 300 KiB limit, demonstrating that quantisation is necessary for this architecture rather than merely cosmetic.

The official checker applies the 300 KiB hard limit to the model activation arena only. Preprocessing and postprocessing would also consume physical SRAM in a real embedded implementation, so a separate conservative streaming design estimate was performed. With one persistent EMA state, one previous-frame state and reusable scratch/input buffers, the combined design estimate is 255.8 KiB, leaving about 44.2 KiB below the nominal budget. This is a feasibility analysis, not an assignment requirement for C/C++ implementation. The project is evaluated through `code.py`, `model.tflite` and the official checker.

8 Inference pipeline and reproducibility

`submission/code.py` implements the exact V11 preprocessing contract with NumPy and maintains EMA and previous-frame state across consecutive chunks. The inference sequence is:

1. load the supplied `.npy` radar tensor and validate its dimensions;
2. compute magnitude, signed I/Q, phase differences and causal states;
3. apply embedded training-only mean and standard-deviation constants;
4. quantise features using the TFLite input scale and zero point;
5. invoke the full-INT8 interpreter and dequantise the output;

6. retain slots above threshold 0.75, de-normalise coordinates, clip to the room and write one JSON object per frame.

The final submission depends only on the protected repository requirements. Training dependencies are separated into `src/requirements-training.txt` and `src/requirements-eda.txt`. The dataset is downloaded through the standard Hugging Face cache using an environment token; no credential or raw dataset file is committed.

The repository preserves the complete experimental history:

- `01_eda_huggingface.ipynb` and `02_boundary_analysis.ipynb` document data quality;
- notebooks `03a`–`03e` define each preprocessing contract;
- `04_baseline_model.ipynb` contains V1–V12 and INT8 conversion;
- `05_v11_stratified_group_cross_validation.ipynb` reports robust out-of-fold estimates;
- `06_edge_optimization_audit.ipynb` measures size, arena, operations, streaming memory and host latency;
- JSON artifacts store splits, normalisation constants, histories, thresholds and evaluation summaries.

Saved notebook outputs permit inspection without re-training all twelve models, while fixed seeds and uniquely named artifacts prevent one experiment from overwriting another.

9 Innovation, limitations and future work

The main technical contribution is the combination of physical signal structure and edge constraints rather than a larger neural network. Phase differences provide an explicit angular cue; EMA residuals expose change; previous-frame context provides a minimal causal tracking signal; and the physical-distance permutation loss aligns training with the official matching radius. These features add no recurrent or custom TFLite operation and preserve a small inference graph.

The most important limitation is acquisition-session generalisation. Two-fold pooled F1 is 0.6867, and the weakest fold shows poor four-person count accuracy. The dataset has only 24 independent recording groups, so increasing the number of frame samples does not create more independent sessions. Moreover, the previous-frame input is a short temporal cue, not an explicit multi-target tracker: it cannot maintain stable identities through long occlusions or crossings.

Future work should prioritise more diverse acquisition sessions, calibration-robust phase features and session-level augmentation rather than simply increasing model width. A lightweight post-training confidence calibration fitted on group-disjoint data may improve count stability. If physical deployment were required, the streaming buffer plan should be implemented and profiled in TFLM on an ESP32-S3, including end-to-end latency and peak SRAM. Such C/C++ deployment validation lies beyond the deliverables of the present assignment.

10 Conclusion

The project progressed from a 4,372-parameter magnitude baseline to a phase-aware, causal and metric-aligned V11 model through twelve documented experiments. The final float model achieves 0.8516 F1 on the selected group-disjoint validation split, and the deployed full-INT8 model retains 0.8339 F1 in only 155.6 KiB. Its 103.9 KiB activation estimate and standard-operation graph satisfy the updated ESP32-S3 screening constraints.

Equally importantly, the report does not equate the model-selection score with guaranteed unseen-session performance. The pooled cross-validation result exposes a remaining domain gap and defines the principal risk for final evaluation. Within the project constraints, V11 offers the strongest measured balance between localisation, counting, reproducibility and edge feasibility; the unsuccessful V12 experiment supports freezing V11 rather than adding unverified complexity immediately before submission.

References

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[2] H. W. Kuhn, "The Hungarian Method for the Assignment Problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

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